



AZ-140

Configure and Operate
Azure Virtual Desktop

Monitor and manage
performance and health



AZ-140 Agenda

Learning Path 1

1. Azure Virtual Desktop Architecture
2. Design the Azure Virtual Desktop architecture
3. Design for user identities and profiles

Learning Path 2

4. Implement and manage networking for AVD
5. Implement and manage storage for AVD
6. Create and configure host pools and session hosts for AVD
7. Create and manage session host image for AVD

Learning Path 3

8. Manage access for AVD
9. Manage security for AVD

△ App Registration
AVD
Microsoft
Tenant

SP Service Principal (ID)
= Enterprise App

Learning Path 4

10. Implement and manage FSLogix
11. Configure user experience settings
12. Install and configure apps on a session host

Learning Path 5

13. Monitor and manage performance and health
14. Plan and implement updates, backups, and disaster recovery



Student 1 →

SP
AVD

→ start SH on demand
scaling plans

SH

Golden Image
→ Go Deploy Lab 7

Lab 1-6

Introduction

After completing this module, you'll be able to:

- Configure log collection and analysis for Azure Virtual Desktop session hosts
- Monitor Azure Virtual Desktop by using Azure Monitor
- Customize Azure Monitor workbooks for Azure Virtual Desktop Insights
- Optimize for capacity and performance
- Monitor Azure Virtual Desktop by using Azure Advisor
- Implement scaling plans in host pools

Configure log collection and analysis for Azure Virtual Desktop session hosts

Monitor and manage Azure Virtual Desktop services



Azure Virtual Desktop uses Azure Monitor for monitoring and alerts like many other Azure services. The service creates activity logs for both user and administrative actions. Each activity log falls under the following categories:

Category	Description
Management Activities	Whether attempts to change Azure Virtual Desktop objects using APIs or PowerShell are successful.
Feed	Whether users can successfully subscribe to workspaces.
Connections	When users initiate and complete connections to the service.
Host registration	Whether a session host successfully registered with the service upon connecting.
Errors	Where users encounter issues with specific activities.
Checkpoints	Specific steps in the lifetime of an activity that were reached.
Agent Health Status	Monitor the health and status of the Azure Virtual Desktop agent installed on each session host.
Network	The average network data for user sessions to monitor for details including the estimated round-trip time.
Connection Graphics	Performance data from the Azure Virtual Desktop graphics stream.
Session Host Management Activity	Management activity of session hosts.
Autoscale	Scaling operations.

Push diagnostics data to your workspace

You push diagnostics data from your Azure Virtual Desktop objects into the Log Analytics for your workspace. You can set up this feature right away when you first create your objects.

To set up Log Analytics for a new object:

1. Sign in to the Azure portal and go to **Azure Virtual Desktop**.
2. Navigate to the object (such as a host pool, application group, or workspace) that you want to capture logs and events for.
3. Select **Diagnostic settings**
4. Select **Add diagnostic setting**. The options shown in the **Diagnostic Settings** page will vary depending on what kind of object you're editing. Remember to enable diagnostics for each Azure Resource Manager object that you want to monitor. Data will be available for activities after diagnostics has been enabled. It might take a few hours after first set-up.
5. Enter a name for your settings configuration, then select **Send to Log Analytics**. The name you use shouldn't have spaces and should conform to [Azure naming conventions](#).
6. Select **Save**.

How to access Log Analytics

Access Log Analytics on a Log Analytics workspace

1. Sign in to the Azure portal.
2. Search for **Log Analytics workspace**.
3. Under **Services**, select **Log Analytics workspaces**.
4. From the list, select the workspace you configured for your Azure Virtual Desktop object.
5. Once in your workspace, select **Logs**. You can filter out your menu list with the Search function.

Access Log Analytics on Azure Monitor

1. Sign in to the Azure portal.
2. Search for and select **Monitor**.
3. Select **Logs**.
4. Follow the instructions in the logging page to set the scope of your query.
5. You're ready to query diagnostics. All diagnostics tables have a "WVD" prefix.

Monitor Azure Virtual Desktop by using Azure Monitor

Monitor and manage Azure Virtual Desktop services



Monitor Azure Virtual Desktop by using Azure Monitor

Before you start using Azure Monitor for Azure Virtual Desktop, you'll need to set up the following things:

- At least one configured Log Analytics Workspace. Use a designated Log Analytics workspace for your Azure Virtual Desktop session hosts to ensure that performance counters and events are only collected from session hosts.
- Enable data collection for the following things in your Log Analytics workspace:
 - Diagnostics from your Azure Virtual Desktop environment
 - Recommended performance counters from your Azure Virtual Desktop session hosts
 - Recommended Windows Event Logs from your Azure Virtual Desktop session hosts
- The data setup process described in this unit is the only one you'll need to monitor Azure Virtual Desktop.
- You can disable all other items sending data to your Log Analytics workspace to save costs.

Access Azure Monitor

Open Azure Monitor for Azure Virtual Desktop

1. Go to the Azure portal.
2. Search and select **Azure Monitor**.
3. Select **Insights Hub** under Insights, then **Azure Virtual Desktop**.
4. Enter Subscription, Resource group, Host pool, and Time range.

Log Analytics settings

You need at least one Log Analytics workspace. Use a designated workspace for Azure Virtual Desktop session hosts.

Customize Azure Monitor workbooks for Azure Virtual Desktop Insights

Monitor and manage Azure Virtual Desktop services



Customize Azure Monitor workbooks

Azure Virtual Desktop Insights is a dashboard built on Azure Monitor Workbooks.

You'll need to set up the following:

- All Azure Virtual Desktop environments you monitor must be based on the latest release of Azure Virtual Desktop that's compatible with Azure Resource Manager.
- At least one configured Log Analytics Workspace. Use a designated Log Analytics workspace for your Azure Virtual Desktop session hosts.
- Enable data collection for the following things in your Log Analytics workspace:
 - Diagnostics from your Azure Virtual Desktop environment.
 - Recommended performance counters from your Azure Virtual Desktop session hosts.
 - Recommended Windows Event Logs from your Azure Virtual Desktop session hosts.

Resource Diagnostic

Resource diagnostic settings

To set your resource diagnostic settings in the configuration workbook:

1. Select the **Resource diagnostic settings** tab in the configuration workbook.
2. Select **Log Analytics workspace** to send Azure Virtual Desktop diagnostics.

Session host data settings

Use Azure Monitor Agent for session hosts. Configure Data Collection Rule to collect performance data and Windows Event Logs.

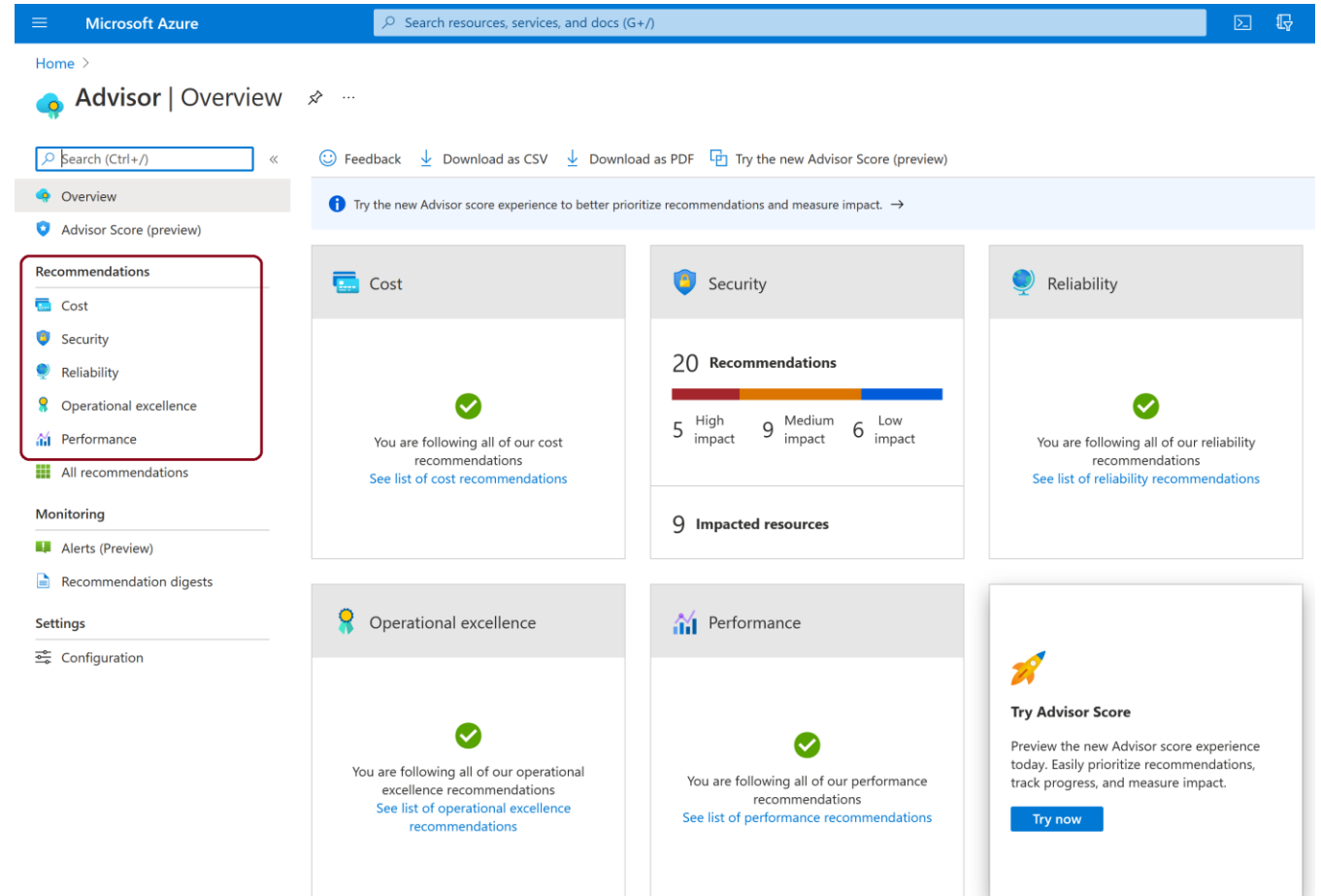
Monitor Azure Virtual Desktop by using Azure Advisor

Monitor and manage Azure Virtual Desktop services



Monitor Azure Virtual Desktop by using Azure Advisor

- Azure Advisor analyzes your configurations and telemetry to offer personalized recommendations to solve common problems.
- You can optimize your Azure resources for reliability, security, operational excellence, performance, and cost.



Getting Started with Azure Advisor

When you open Azure Advisor, you'll see:

- Cost
- Security
- Reliability
- Operational Excellence
- Performance

Microsoft Azure

Search resources, services, and docs (G+)

Home > Advisor

Advisor | Security

Search (Ctrl+ /)

Feedback Download as CSV Download as PDF Try the new Advisor Score (preview)

Subscriptions: Azure Pass - Sponsorship Active No grouping

Prevent, detect and respond to threats with increased visibility, go to Security Center →

Total recommendations: 20

Recommendations by impact: 5 High impact, 9 Medium impact, 6 Low impact

Impacted resources: 9

Security alerts: 0

Learn more: What is Security Center, Explore Security Center Recommendations

To see more about Secure Score and Security recommendations, visit Security Center

Impact	Description	Impacted resources	Last updated
High	Management ports of virtual machines should be protected with just-in-time network access control	1 Virtual machine	4/25/2021, 5:34 PM
High	Disk encryption should be applied on virtual machines	5 Virtual machines	4/25/2021, 5:34 PM
High	All network ports should be restricted on network security groups associated to your virtual machine	1 Virtual machine	4/25/2021, 5:34 PM
High	Windows web servers should be configured to use secure communication protocols	4 Virtual machines	4/25/2021, 5:34 PM
High	There should be more than one owner assigned to your subscription	1 Subscription	4/25/2021, 5:34 PM
Medium	Storage accounts should restrict network access using virtual network rules	2 Storage Accounts	4/25/2021, 5:34 PM
Medium	Management ports should be closed on your virtual machines	1 Virtual machine	4/25/2021, 5:34 PM
Medium	Network traffic data collection agent should be installed on Windows virtual machines	4 Virtual machines	4/25/2021, 5:34 PM

How to resolve Azure Advisor recommendations

Monitor and manage Azure Virtual Desktop services



How to resolve Azure Advisor recommendations

Recommendations to be resolved:

- No validation environment enabled.
- Not enough production (non-validation) environments enabled.
- Not enough links are unblocked to successfully implement your VM.

The screenshot shows the Microsoft Azure Advisor | Operational Excellence page. The left sidebar contains navigation links: Overview, Recommendations, Monitoring, and Settings. The Recommendations section is active, showing a list of recommendations. The main content area displays a summary of recommendations by impact (High, Medium, Low) and a table of recommendations. A red box highlights the recommendation 'No validation environment enabled' with a Medium impact.

IMPACT	DESCRIPTION	POTENTIAL BENEFITS	IMPACTED RESOURC...	UPDATED AT
High	Enforce 'Audit that do not use managed disks' using Azure Policy	This specific policy audits that do not use managed disks.	1 Subscription	8/12/2020, 5:02:34 AM
Medium	No validation environment enabled	Ensure business continuity through service deployments	1 Host Pool	8/12/2020, 11:03:25 AM
Medium	Enforce 'Inherit a tag from the resource group' using Azure Policy	This specific policy allows for resources to inherit tags from the parent resource group.	1 Subscription	8/12/2020, 5:01:46 AM
Medium	Enforce 'Allowed virtual machine SKUs' using Azure Policy	This specific policy restricts the virtual machine SKUs that your organization can deploy.	1 Subscription	8/12/2020, 5:02:26 AM
Medium	Enforce 'Add or replace a tag on resources' using Azure Policy	This specific policy allows for resources to be tagged or for tags to be replaced.	1 Subscription	8/12/2020, 5:01:34 AM
Medium	Enforce 'Allowed locations' using Azure Policy	This specific policy restricts the available locations where your organization can deploy new resources.	1 Subscription	8/12/2020, 5:01:38 AM
Low	Create an Azure service health alert	Get notified when Azure service issues affect you	1 Subscription	8/12/2020, 9:50:43 AM

No validation environment enabled

Home >

0618HP | Properties

Host pool

Search (Ctrl+ /)

<<

Save

Discard

Download template

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Settings

Properties

Locks

Export template

Manage

Application groups

Session hosts

Change subscription

Location

East US

Name

0618HP

Configuration

Personal desktop

Yes

No

Validation environment ⓘ

No

Yes

Friendly name

Description

Created through the WVD extension

Diagnose graphics performance issues

Monitor and manage Azure Virtual Desktop services



Diagnose graphics performance issues

To diagnose experience quality issues with your remote sessions, counters have been provided under the RemoteFX Graphics section of Performance Monitor.

Find a remote session name

You'll need your remote session name to identify the graphics performance counters.

1. Open the Windows command prompt from your remote session.
2. Run the **qwinsta** command and find your session name.
 - If your session is hosted in a multi-session virtual machine (VM): Your instance of each counter is suffixed by the same number that suffixes your session name, such as "rdp-tcp 37."
 - If your session is hosted in a VM that supports virtual Graphics Processing Units (vGPU): Your instance of each counter is stored on the server instead of in your VM. Your counter instances include the VM name instead of the number in the session name, such as "Win8 Enterprise VM."

Addressing low frame rate, random stalls, and high input latency

First check the Output Frames/Second counter. It measures the number of frames made available to the client. If this value is less than the Input Frames/Second counter, frames are being skipped.

To identify the bottleneck, use the Frames Skipped/Second counters.

There are three types of Frames Skipped/Second counters:

- Frames Skipped/Second (Insufficient **Server** Resources)
- Frames Skipped/Second (Insufficient **Network** Resources)
- Frames Skipped/Second (Insufficient **Client** Resources)

A high value for any of the Frames Skipped/Second counters implies that the problem is related to the resource the counter tracks.

If the Output Frames/Second counter matches the Input Frames/Second counter, yet you still notice unusual lag or stalling, Average Encoding Time may be the culprit. Encoding is a synchronous process that occurs on the server in the single-session (vGPU) scenario and on the VM in the multi-session scenario. Average Encoding Time should be under 33 ms.

Implement scaling plans in host pools

Monitor and manage Azure Virtual Desktop services



Steps to Create a Scaling Plan

Before creating your first scaling plan, you'll need to assign the Desktop Virtualization Power On Off Contributor RBAC role to the Azure Virtual Desktop service principal with your Azure subscription as the assignable scope.

To create a scaling plan using the portal:

1. Sign in to the Azure portal.
2. In the search bar, type *Azure Virtual Desktop* and select the matching service entry.
3. Select **Scaling Plans**, then select **Create**.
4. In the **Basics** tab, look under Project details and select the name of the subscription you'll assign the scaling plan to.
5. If you want to make a new resource group, select **Create new**. If you want to use an existing resource group, select its name from the drop-down menu.
6. Enter a name for the scaling plan into the **Name** field.
7. For **Region**, select a region for your scaling plan.
8. For **Time zone**, select the time zone you'll use with your plan.
9. For **Host pool type**, select the type of host pool that you want your scaling plan to apply to.
10. In **Exclusion tags**, enter a tag name for VMs you don't want to include in scaling operations.
11. Select **Next**, which should take you to the Schedules tab.

Schedules let you define when autoscale turns VMs on and off throughout the day. The schedule parameters are different based on the Host pool type you chose for the scaling plan.

Optimize capacity and performance

Monitor and manage Azure Virtual Desktop services



Optimize capacity and performance

Use the Performance Diagnostics tool to identify and troubleshoot performance issues on your Azure virtual machine (VM) in one of two modes:

- **Continuous diagnostics (preview)** collects data at five-second intervals and reports actionable insights about high resource usage every five minutes.
- **On-demand diagnostics** helps you troubleshoot an ongoing performance issue with more in-depth data, insights, and recommendations based on data collected at a single point in time.

Performance Diagnostics stores all insights and reports in a storage account, which you can configure for short data retention to minimize costs.

Run Performance Diagnostics directly from the Azure portal, where you can also review insights and a report on various logs, rich configuration, and diagnostics data. We recommend that you run Performance Diagnostics and review the insights and diagnostics data before you contact Microsoft Support.

Knowledge check

Monitor and manage Azure Virtual Desktop services



Knowledge check

A system administrator is tasked with diagnosing experience quality issues during Remote Desktop Protocol (RDP) sessions. They have identified the remote session name and are now looking to collect the RemoteFX Graphics performance counters. What should they do next?

Choices:

1. They should run the qwinsta command again.
2. They should restart the remote session.
3. They should select Start > Administrative Tools > Performance Monitor, expand Monitoring Tools, select Performance Monitor, and then select Add.

Knowledge check

A system administrator is tasked with setting up diagnostics for an Azure Virtual Desktop workspace. They notice that diagnostics are not enabled and an error message appears. What should the administrator do to enable the supported diagnostics tables?

Choices:

1. They should disable the Azure Virtual Desktop workspace and re-enable it
2. They should ignore the error message and proceed with the setup
3. They should enable the following supported diagnostics tables: Management Activities Feed, Errors, Checkpoints

Knowledge check

A system administrator needs to monitor activities on Azure Virtual Desktop. They have set up diagnostics and named their settings configuration according to Azure naming conventions. What should they do next to access the Log Analytics workspace?

Choices:

1. They should search for 'Azure Monitor' in the Azure portal and select 'Logs' from the list.
2. They should wait for a few hours for the data to be available and then directly access the data from the diagnostics setup.
3. They should sign in to the Azure portal, search for 'Log Analytics workspace', select it under Services, and choose the workspace they configured for their Azure Virtual Desktop object.

Knowledge check

An IT professional is tasked with setting up Azure Virtual Desktop Insights to monitor their Azure Virtual Desktop environments. What is the minimum Azure role based access control (RBAC) roles they need to have assigned?

Choices:

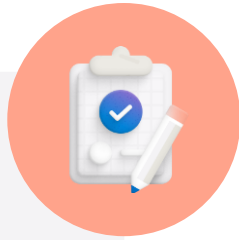
1. Desktop Virtualization Writer and Log Analytics Writer
2. Desktop Virtualization Reader and Log Analytics Writer
3. Desktop Virtualization Reader and Log Analytics Reader

Summary

Monitor and manage Azure Virtual Desktop services



Summary



What you learned:

- Configure log collection and analysis for Azure Virtual Desktop session hosts
- Monitor Azure Virtual Desktop by using Azure Monitor
- Customize Azure Monitor workbooks for Azure Virtual Desktop Insights
- Optimize for capacity and performance
- Monitor Azure Virtual Desktop by using Azure Advisor
- Implement scaling plans in host pools